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Can changes in eye appearance be used to predict high blood pressure?

Hypertension can display morphological symptoms that include changes in eye appearance and color from hemorrhaging blood vessels. The goal of this experiment was to determine whether eye appearance can specifically be used to predict high blood pressure. Using an adult male with hypertension as a test subject, eye images and blood pressure readings were collected together over 4 weeks. Eye images were taken with an Iphone camera set at a fixed orientation, along with corresponding readings in mmHg using a blood pressure monitor. Diet and physical activity were noted. A threshold of 130/80 mmHg was used to classify readings as high (1) or low (0). The images and the pressure data were merged and fed into a neural network for binary (high/low) classification. The neural network had 87.5 - 91.7% accuracy after being trained with 67 images. On the test set of 8 images, it achieved > 60% accuracy (5 out of 8 correctly classified). The network seems capable of detecting a change in blood pressure state from features related to eye appearance. This shows promising results supporting the initial hypothesis. However there is a risk that the network may have overfit the training data due to fewer examples. Further research includes collecting additional higher quality images to improve performance, as well as predicting numeric outputs instead of binary classification. The model can be implemented in a mobile app that classifies blood pressure from images.